

# ONCOGENE ADVANCED DIAGNOSTICS LABORATORY

Multi-Cancer Early Detection (MCED) Next-Gen Liquid Biopsy Panel

CLIA ID: 24D2094819  
CAP Accredited: #8491028  
Clinical Protocol: MCED-v2.4

## PATIENT IDENTIFICATION

**Legal Name:** Arthur Pendelton Hayes  
**Date of Birth:** 1961-04-14  
**Medical Record #:** MRN-7840192  
**Gender / Sex:** Male (Assigned at Birth)  
**Residential Address:** 1482 Highland Parkway, St. Paul, MN 55116  
**Contact Phone:** (651) 555-0194

## SPECIMEN & ORDER INFORMATION

**Accession ID:** ACC-2026-MCED-9184  
**Collection Date:** 2026-08-28 09:15 CDT  
**Received Date:** 2026-08-29 07:45 CDT  
**Report Date:** 2026-09-02 14:30 CDT  
**Ordering Physician:** Dr. Evelyn Montgomery, MD  
**Clinical Facility:** Midwest Oncology Associates - Main Campus

## OVERALL CANCER SIGNAL STATUS: **SIGNAL DETECTED**

**Primary Predicted Signal Origin (CSO 1):** Colorectal (Probability Score: 0.84)  
**Secondary Predicted Origin (CSO 2):** Gastric / Gastroesophageal (Probability Score: 0.11)  
**Estimated ctDNA Tumor Fraction (%):** **3.8% (High Positive Circulating Load)**  
**Signal Confidence Classification:** Tier 1 - High Specificity Genomic & Epigenetic Concordance

## CLINICAL INTERPRETATION & ACTIONABLE SUMMARY

A circulating cell-free DNA (cfDNA) cancer signal was DETECTED in the plasma specimen. Machine learning pattern analysis of high-depth targeted somatic sequencing combined with whole-genome bisulfite methylation profiling indicates a primary tissue of origin localization of Colorectal with a calibrated probability index of 0.84.

Multiple pathogenic somatic driver mutations in colorectal-associated lineage pathways were confirmed at high analytical confidence with an overall estimated circulating tumor fraction of 3.8%. Diagnostic workup including high-resolution targeted colonoscopy / cross-sectional imaging (CT Abdomen/Pelvis with IV contrast) is recommended.

Electronically Verified & Released By:

**Dr. Nathan K. Sterling, MD, PhD, FCAP**

Medical Director of Molecular Pathology | NPI: 1948201948

Diagnostic Facility:

OncoGene Diagnostics - Central Laboratory

9200 Medical Center Way, Suite 400, Minneapolis, MN



PATIENT: ARTHUR PENDELTON HAYES | DOB: 1961-04-14 | MRN: MRN-7840192  
ACCESSION: ACC-2026-MCED-9184 | PANEL: MCED Next-Gen Liquid Biopsy Panel (ctDNA)

## SECTION 1: TARGETED SOMATIC GENOMIC ALTERATIONS (ctDNA NGS)

| Gene Symbol                                                                                                               | DNA / Amino Acid Alteration       | Target Exon | Variant Fraction (VAF %) | Clinical Significance / Pathogenicity Tier     | Status   |
|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------|--------------------------|------------------------------------------------|----------|
| KRAS                                                                                                                      | c.35G>A (p.Gly12Asp / G12D)       | Exon 2      | 2.4%                     | Tier I - Pathogenic (Colorectal / GI Lineage)  | DETECTED |
| APC                                                                                                                       | c.4348C>T (p.Arg1450Ter / R1450*) | Exon 16     | 3.1%                     | Tier I - Pathogenic (Truncating / Wnt Pathway) | DETECTED |
| TP53                                                                                                                      | c.818G>A (p.Arg273His / R273H)    | Exon 8      | 1.9%                     | Tier I - Pathogenic (DNA Binding Domain)       | DETECTED |
| Non-Detected Genes: BRAF, NRAS, PIK3CA, EGFR, ERBB2, MET, RET, ALK, ROS1 (Below analytical sensitivity limit of 0.1% VAF) |                                   |             |                          |                                                |          |

## SECTION 2: EPIGENETIC METHYLATION PATTERNS & CIRCULATING PROTEIN MARKERS

| Biomarker / Assay Target                 | Biological Modality | Quantitative Index / Value | Reference Cutoff Baseline  | Result Interpretation     |
|------------------------------------------|---------------------|----------------------------|----------------------------|---------------------------|
| SEPT9 Promoter Methylation (sept9_index) | Targeted Bisulfite  | 0.79                       | < 0.20 (Negative Baseline) | ELEVATED HYPERMETHYLATION |
| SDC2 Promoter Methylation (sdc2_index)   | Targeted Bisulfite  | 0.68                       | < 0.25 (Negative Baseline) | ELEVATED HYPERMETHYLATION |
| NDRG4 Promoter Methylation (ndrg4_index) | Targeted Bisulfite  | 0.72                       | < 0.30 (Negative Baseline) | ELEVATED HYPERMETHYLATION |
| Serum Carcinoembryonic Antigen (CEA)     | Immunoassay ECLIA   | 12.4 ng/mL                 | < 3.0 ng/mL (Non-Smoker)   | ELEVATED CIRCULATING      |

#### ASSAY METHODOLOGY, ANALYTICAL SENSITIVITY & REGULATORY DISCLAIMERS

Methodology: High-throughput sequencing of circulating cell-free DNA (cfDNA) isolated from peripheral whole blood EDTA tubes.  
Analytical Performance: Somatic SNV/Indel limit of detection (LOD) = 0.1% VAF at 50,000x raw depth. Methylation analytical specificity = 99.2%.  
Regulatory Notice: Developed and validated pursuant to CLIA regulations. This test is intended to be used in conjunction with clinical diagnostic guidelines.